

COVID-19 Trends

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Introduction

On 31st December 2019, reports of a contagious respiratory illness of unknown cause surfaced in Wuhan, China. Roughly one month later, on 30th January 2020, the World Health Organization (WHO) declared the outbreak as a public health emergency of international concern, scaling this assessment to pandemic status on 11th March 2020.¹

The cause of this outbreak was discovered to be a coronavirus, specifically the SARS-CoV-2 virus, afflicting people with the disease COVID-19 (**CO**rona**VI**rus **D**isease). Transmission of the virus mainly occurs via airborne droplets containing viral particles, though any form of contact with viral particles can lead to infection.² Symptoms of the disease range from asymptomatic to deadly, most commonly afflicting patients with: fever, sore throat, cough, fatigue, breathing difficulty, loss of smell and loss of taste.³

In order to better understand how the pandemic progressed, analysis of data obtained during the pandemic is key.

Aims

- To develop an understanding of the data analysis workflow using R.
- Gain insights into the progression of the COVID-19 pandemic, particularly:
 - Which countries have the greatest number of positive cases relative to the number of tests conducted in the early stages of the pandemic?
 - How did the UK handle the initial months of the pandemic?

Dataset

A publicly available dataset was sourced from Kaggle.

Data Dictionary

Column	Description
Date	Date
Continent	Name of continent (North America, Europe, ...)
Country_Code	Country code (UK, US, CA, ...)
Country	Country name
Province_State	State/Province names ("All States" where state/province data not available)
positive	Cumulative number of reported positive cases
active	Number of active cases on that day

Column	Description
hospitalized	Cumulative number of reported hospitalized cases
hospitalizedCurr	Number of active hospitalized cases that day
recovered	Cumulative number of recovered cases
death	Cumulative number of deaths
total_tested	Cumulative number of tests conducted
daily_tested	Number of tests conducted on the day. If daily data unavailable, daily tested averaged across days in between
daily_positive	Number of positive cases reported on the day. If daily data unavailable, daily positive averaged across days in between

Loading Libraries and the Dataset

```
# Loading libraries
library(conflicted)
library(tidyverse)

## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.2.0      v readr      2.2.0
## v forcats    1.0.1      v stringr    1.6.0
## v ggplot2    4.0.2      v tibble     3.3.1
## v lubridate  1.9.5      v tidyr      1.3.2
## v purrr      1.2.1

# Resolve filter conflict
conflicts_prefer(dplyr::filter)

## [conflicted] Will prefer dplyr::filter over any other package.

# Loading dataset
covid_df <- read_csv("covid19.csv")

## Rows: 10903 Columns: 14
## -- Column specification -----
## Delimiter: ","
## chr (5): Date, Continent, Country_Code, Country, Province_State
## dbl (9): positive, hospitalized, recovered, death, total_tested, active, hos...
##
## i Use 'spec()' to retrieve the full column specification for this data.
## i Specify the column types or set 'show_col_types = FALSE' to quiet this message.

# Dimension of the dataframe
dim(covid_df)

## [1] 10903      14
```

```
# Column names
```

```
vector_cols <- names(covid_df)
vector_cols
```

```
## [1] "Date"          "Continent"      "Country_Code"   "Country"
## [5] "Province_State" "positive"        "hospitalized"    "recovered"
## [9] "death"         "total_tested"   "active"          "hospitalizedCurr"
## [13] "daily_tested"   "daily_positive"
```

```
# First six observations
```

```
knitr::kable(head(covid_df))
```

Date	Continent	Country_Code	Country	Province_State	positive	hospitalized	recovered	death	total_tested	active	hospitalizedCurr	daily_tested	daily_positive
20/01/2020	Asia	KR	South Korea	All States	1	0	0	0	4	0	0	0	0
22/01/2020	North America	US	United States	All States	1	0	0	0	1	0	0	0	0
22/01/2020	North America	US	United States	Washington	1	0	0	0	1	0	0	0	0
23/01/2020	North America	US	United States	All States	1	0	0	0	1	0	0	0	0
23/01/2020	North America	US	United States	Washington	1	0	0	0	1	0	0	0	0
24/01/2020	Asia	KR	South Korea	All States	2	0	0	0	27	0	0	5	0

```
# Global view of the dataset
```

```
glimpse(covid_df)
```

```
## Rows: 10,903
## Columns: 14
## $ Date          <chr> "20/01/2020", "22/01/2020", "22/01/2020", "23/01/2020~
## $ Continent     <chr> "Asia", "North America", "North America", "North Amer~
## $ Country_Code  <chr> "KR", "US", "US", "US", "US", "KR", "US", "US", "AU",~
## $ Country       <chr> "South Korea", "United States", "United States", "Uni~
## $ Province_State <chr> "All States", "All States", "Washington", "All States~
## $ positive      <dbl> 1, 1, 1, 1, 1, 2, 1, 1, 4, 0, 3, 0, 0, 0, 1, 0, 1,~
## $ hospitalized  <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,~
## $ recovered     <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,~
## $ death        <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,~
## $ total_tested  <dbl> 4, 1, 1, 1, 1, 27, 1, 1, 0, 0, 0, 0, 0, 0, 0, 0, 3~
## $ active       <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,~
## $ hospitalizedCurr <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,~
## $ daily_tested  <dbl> 0, 0, 0, 0, 0, 5, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,~
## $ daily_positive <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,~
```

The dataset `covid_df` contains 10903 rows and 14 columns. All data types and column names are sensible.

Isolating Required Rows

```
# isolating country level data
covid_df_all_states <- covid_df %>%
  filter(Province_State == "All States") %>%
  select(-Province_State)

knitr::kable(head(covid_df_all_states))
```

Date	Continent	Country_Code	Country	positive	hospitalized	recovered	deaths	total_tested	active	hospitalizedCurr	daily_tested	daily_positive
20/01/2020	Asia	KR	South Korea	1	0	0	0	4	0	0	0	0
22/01/2020	North America	US	United States	1	0	0	0	1	0	0	0	0
23/01/2020	North America	US	United States	1	0	0	0	1	0	0	0	0
24/01/2020	Asia	KR	South Korea	2	0	0	0	27	0	0	5	0
24/01/2020	North America	US	United States	1	0	0	0	1	0	0	0	0
25/01/2020	Oceania	AU	Australia	4	0	0	0	0	0	0	0	0

- We can remove `Province_State` without losing information because after the filtering step this column only contains the value "All States".

Isolating Required Columns

The dataset contains a mixture of data collected on a daily basis and cumulative frequency counts. We must separate these such that we are not comparing daily data with cumulative data (this will lead to inaccurate conclusions otherwise).

```
# isolating daily data
covid_df_all_states_daily <- covid_df_all_states %>%
  select(Date, Country, active, hospitalizedCurr, daily_tested, daily_positive)

knitr::kable(head(covid_df_all_states_daily))
```

Date	Country	active	hospitalizedCurr	daily_tested	daily_positive
20/01/2020	South Korea	0	0	0	0
22/01/2020	United States	0	0	0	0
23/01/2020	United States	0	0	0	0
24/01/2020	South Korea	0	0	5	0

Date	Country	active	hospitalizedCurr	daily_tested	daily_positive
24/01/2020	United States	0	0	0	0
25/01/2020	Australia	0	0	0	0

```
covid_df_all_states_cumulative <- covid_df_all_states %>%
  select(Date, Country, positive, hospitalized, recovered, death, total_tested)

knitr::kable(head(covid_df_all_states_cumulative))
```

Date	Country	positive	hospitalized	recovered	death	total_tested
20/01/2020	South Korea	1	0	0	0	4
22/01/2020	United States	1	0	0	0	1
23/01/2020	United States	1	0	0	0	1
24/01/2020	South Korea	2	0	0	0	27
24/01/2020	United States	1	0	0	0	1
25/01/2020	Australia	4	0	0	0	0

Summarizing the Available Daily Data

```
covid_df_all_states_daily_sum <- covid_df_all_states_daily %>%
  group_by(Country) %>%
  summarize(
    tested = sum(daily_tested),
    positive = sum(daily_positive),
    active = sum(active),
    hospitalized = sum(hospitalizedCurr)) %>%
  arrange(-tested)

covid_top_10 <- head(covid_df_all_states_daily_sum, 10)
knitr::kable(covid_top_10)
```

Country	tested	positive	active	hospitalized
United States	17282363	1877179	0	0
Russia	10542266	406368	6924890	0
Italy	4091291	251710	6202214	1699003
India	3692851	60959	0	0
Turkey	2031192	163941	2980960	0
Canada	1654779	90873	56454	0
United Kingdom	1473672	166909	0	0
Australia	1252900	7200	134586	6655
Peru	976790	59497	0	0
Poland	928256	23987	538203	0

We can now begin to analyse the data to answer our initial question: **Which countries have the greatest number of positive cases relative to the number of tests conducted in the early stages of the pandemic?** To do this, we can find the ratio of positive tests to total number of cases conducted for the countries with the greatest number of tests conducted.

```

countries <- covid_top_10$Country
tested_cases <- covid_top_10$tested
positive_cases <- covid_top_10$positive
active_cases <- covid_top_10$active
hospitalized_cases <- covid_top_10$hospitalized

names(tested_cases) <- countries
names(positive_cases) <- countries
names(active_cases) <- countries
names(hospitalized_cases) <- countries

positive_tested_ratios <- (positive_cases / tested_cases)

covid_top_10 <- covid_top_10 %>%
  mutate(
    positive_tested_ratio = positive / tested) %>%
  arrange(-positive_tested_ratio)

knitr::kable(covid_top_10)

```

Country	tested	positive	active	hospitalized	positive_tested_ratio
United Kingdom	1473672	166909	0	0	0.1132606
United States	17282363	1877179	0	0	0.1086182
Turkey	2031192	163941	2980960	0	0.0807117
Italy	4091291	251710	6202214	1699003	0.0615234
Peru	976790	59497	0	0	0.0609107
Canada	1654779	90873	56454	0	0.0549155
Russia	10542266	406368	6924890	0	0.0385466
Poland	928256	23987	538203	0	0.0258409
India	3692851	60959	0	0	0.0165073
Australia	1252900	7200	134586	6655	0.0057467

We can see the UK, USA and Turkey seem to have the greatest ratios, perhaps indicating a significant rate of spread developing in these nations at the start of the pandemic.

Relevant data from these countries are added to a matrix:

```

united_kingdom <- c(0.11, 1473672, 166909, 0, 0)
united_states <- c(0.10, 17282363, 1877179, 0, 0)
turkey <- c(0.08, 2031192, 163941, 2980960, 0)

covid_mat <- rbind(united_kingdom, united_states, turkey)
colnames(covid_mat) <- c("Ratio", "tested", "positive", "active", "hospitalized")

knitr::kable(covid_mat)

```

	Ratio	tested	positive	active	hospitalized
united_kingdom	0.11	1473672	166909	0	0
united_states	0.10	17282363	1877179	0	0
turkey	0.08	2031192	163941	2980960	0

Creating a Summary List

To pack together everything into one place, a summary list is created which can be used to store the various data structures created in this workspace.

```
question <- "Which countries had the greatest ratio of positive cases to tests conducted (circa May/June)"

answer <- c(covid_mat[1:3, 1])

datasets <- list(
  original = covid_df,
  allstates = covid_df_all_states,
  daily = covid_df_all_states_daily,
  cumulative = covid_df_all_states_cumulative,
  top_10 = covid_top_10
)

matrices <- list(covid_mat)
vectors <- list(vector_cols, countries)

data_structure_list <- list("dataframe" = datasets, "matrix" = matrices, "vector" = vectors)

covid_analysis_list <- list(question, answer, data_structure_list)

covid_analysis_list[[2]] # display answer object contained in the list
```

```
## united_kingdom  united_states      turkey
##              0.11              0.10      0.08
```

Comparison to Cumulative Data

To conclude the report, it is interesting to compare the manual summed daily data to the cumulative data.

```
covid_df_all_states_cumulative_summary <- covid_df_all_states_cumulative %>%
  group_by(Country) %>%
  summarize(
    date = max(Date),
    tested = max(total_tested),
    death = max(death),
    hospitalized = max(hospitalized),
    recovered = max(recovered)) %>%
  arrange(-tested)

knitr::kable(head(covid_df_all_states_cumulative_summary, 20))
```

Country	date	tested	death	hospitalized	recovered
United States	31/05/2020	16936891	98536	0	0
Russia	31/05/2020	10643124	4693	0	171883
Spain	30/04/2020	4063843	0	0	0
Germany	29/03/2020	3952971	0	0	0
Italy	31/05/2020	3878739	33415	0	157507

Country	date	tested	death	hospitalized	recovered
India	31/05/2020	3837207	0	0	0
Brazil	28/05/2020	3129744	0	0	0
Turkey	31/05/2020	2039194	4540	0	127973
United Arab Emirates	29/04/2020	1924234	0	0	0
Canada	31/05/2020	1665831	3682	0	48892
Australia	31/05/2020	1476416	102	0	6619
United Kingdom	31/03/2020	1460517	33186	0	0
Peru	31/05/2020	1058874	0	0	0
Iran	31/05/2020	935894	0	0	0
Poland	31/05/2020	931520	1065	0	11449
South Korea	31/05/2020	921391	271	0	10422
Venezuela	31/03/2020	832526	0	0	0
France	26/04/2020	831174	0	0	0
Kazakhstan	31/05/2020	812881	0	0	0
South Africa	31/05/2020	701883	0	0	0

On the whole, the figures are similar, but with slight discrepancies. Why is there a slight difference? This is not entirely clear since this was an incredibly complicated period of time for humanity. It is likely there was a decent amount of missing daily data for each country for a variety of reasons - perhaps some countries could not begin accurate reporting of daily cases until later for example.

Additionally, looking at the cumulative data, we see inconsistent dates for the most recent cumulative data for each country. This dataset was gathered during the escalation of the pandemic which could potentially be one reason why many countries do not seem to be keeping timely updates in the reported figures. For some countries, the dataset provides more recent cumulative data whilst others lag behind by weeks.

We can therefore expect the countries with the most recent dates to have a greater incidence of cases and tests conducted. There are exceptions to this of course since the spread of the virus was far more rapid in some regions/countries compared to others due to a number of factors: geography, politics, social culture *etc.*

Conclusions and Insights into the UK's initial response to the pandemic

The UK was notably hit hard.

From the cumulative data, we see 1460517 tests were conducted up to 13th May 2020, with 33186 reported deaths. This is the largest ratio of deaths to tests conducted (at 0.0227220908760391) for any country in the top 20 by most tests conducted, suggesting poor mobilization of resources.

Additionally, from the daily data summary, it was found that the UK had a positive-test to total-tests ratio of 0.11 which is significantly larger than many of the top 10 countries with the greatest number of tests during the early stages of the pandemic.

Both of these points suggest a poorly managed response from the UK government and general populace. This is corroborated by reports conducted as part of the UK's COVID-19 inquiry.⁴